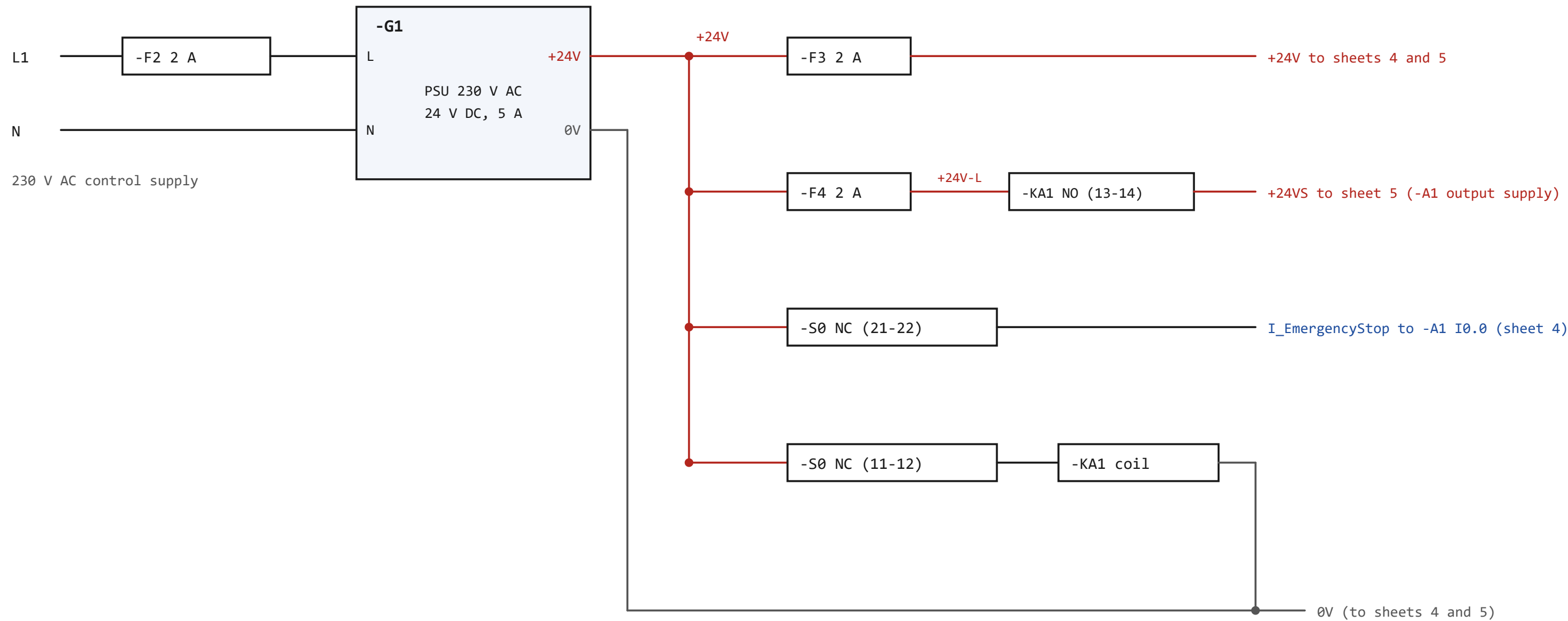
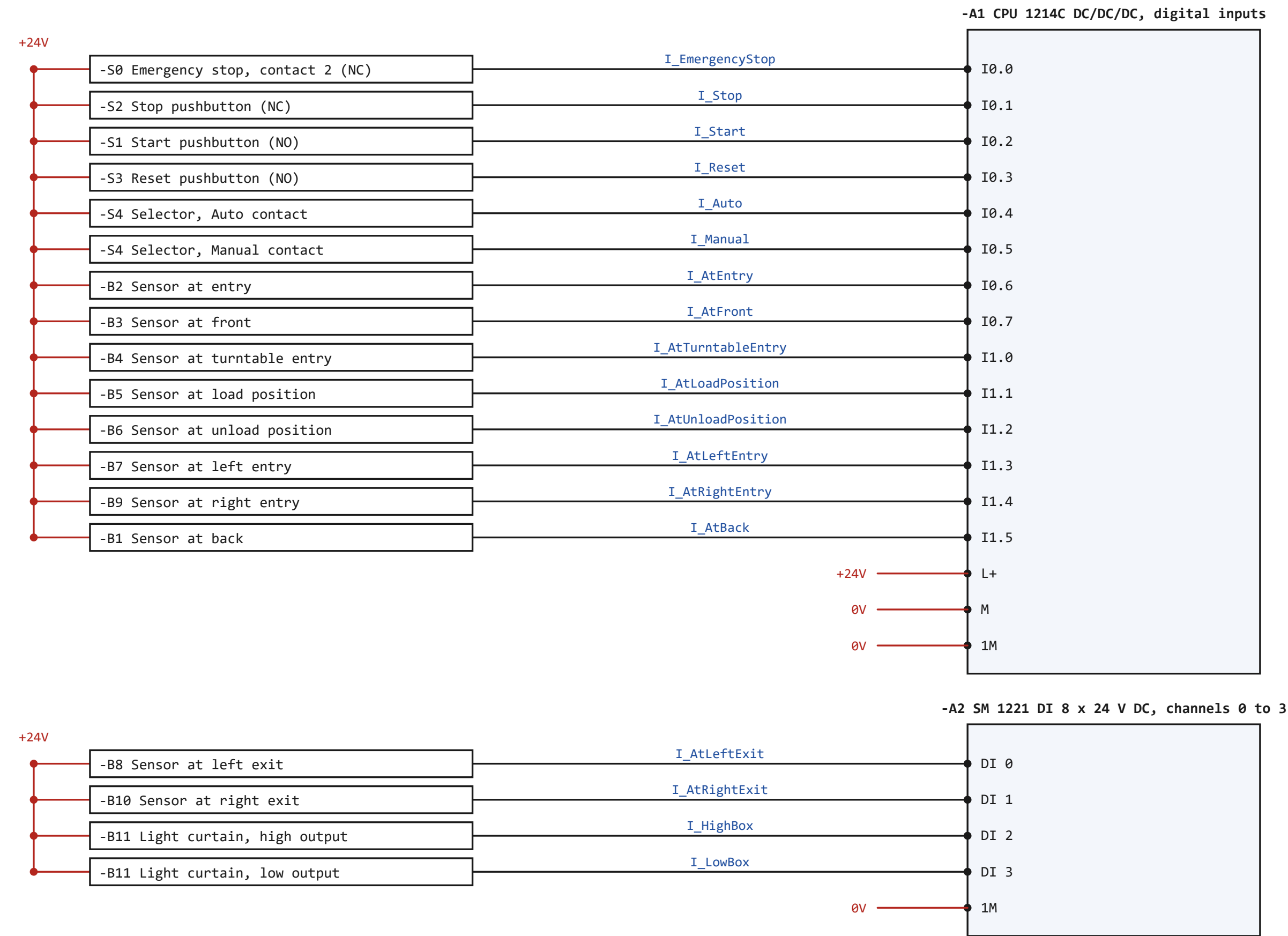


Sheet 3: Control supply 24 V DC and emergency stop



-S0 is an emergency stop with two NC contacts. Contact 1 feeds -KA1, contact 2 is the PLC input.
-KA1 is a standard relay. A certified safety relay is required on a real machine.

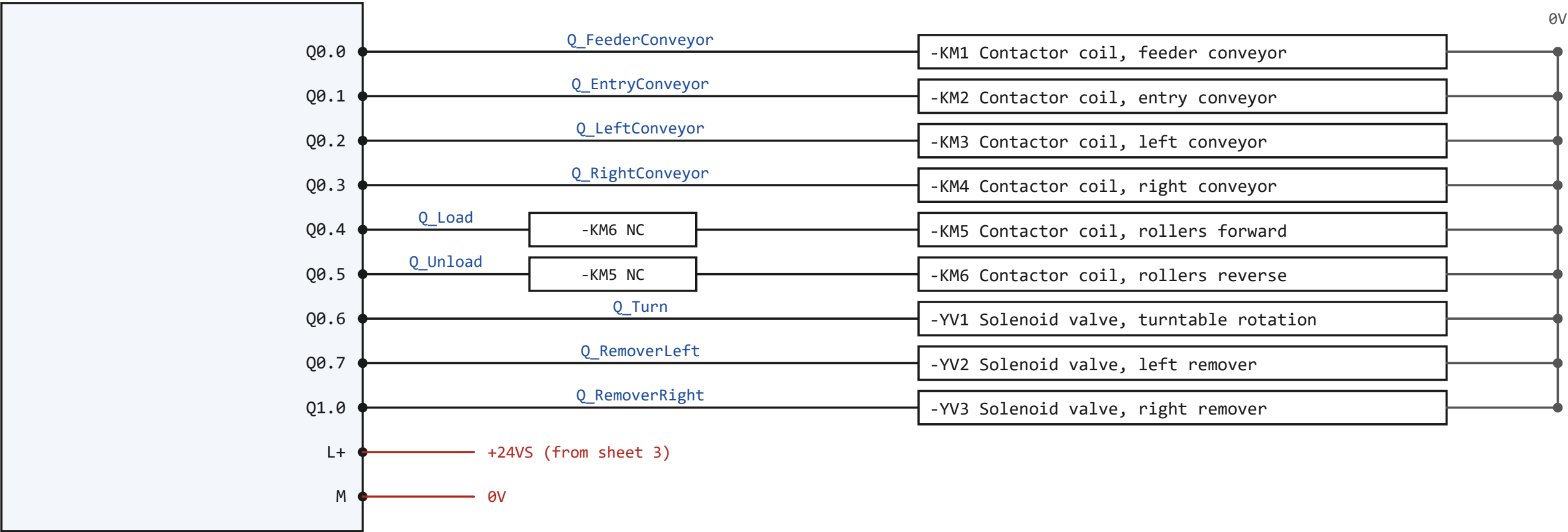
Sheet 4: PLC inputs



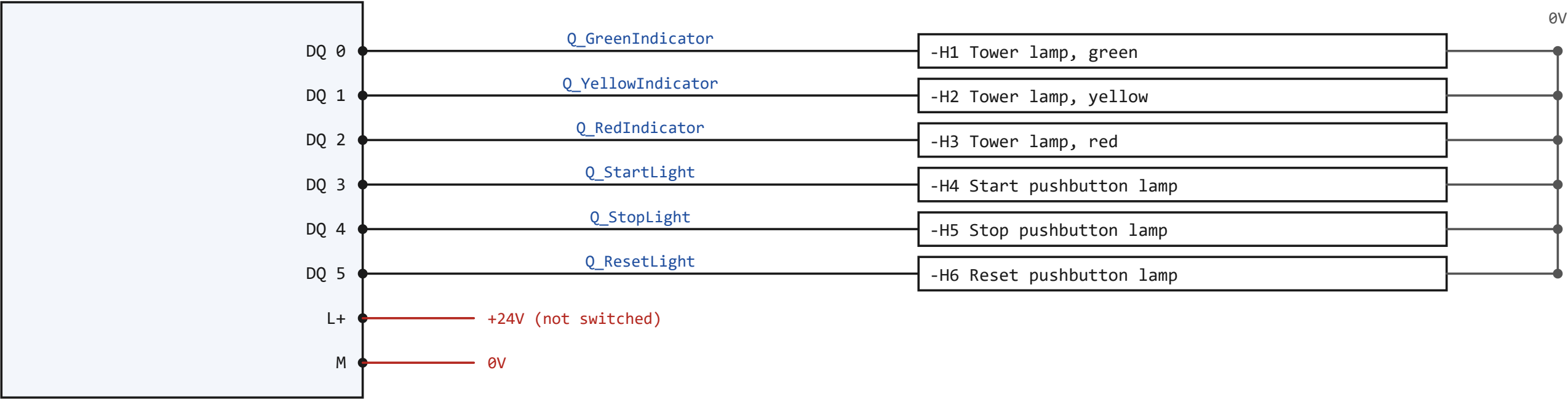
Sensors -B1 to -B11: PNP, three-wire (brown +24V, blue 0V, black signal). Only the +24V and signal wires are drawn.
PLC terminal names (L+, M, 1M) must be checked against the Siemens S7-1200 System Manual. Spare channels are not drawn.

Sheet 5: PLC outputs

-A1 CPU 1214C DC/DC/DC, digital outputs



-A3 SM 1222 DQ 8 x 24 V DC, channels 0 to 5



-KM5 and -KM6 are interlocked by NC auxiliary contacts. Motor power circuits are on sheets 1 and 2.
-A1 outputs are supplied from +24VS, cut by -KA1 on emergency stop. -A3 lamps are supplied from +24V and stay available.
Not wired (simulation only): Q_Emit and Q_Counter. PLC terminal names must be checked against the Siemens S7-1200 System Manual.